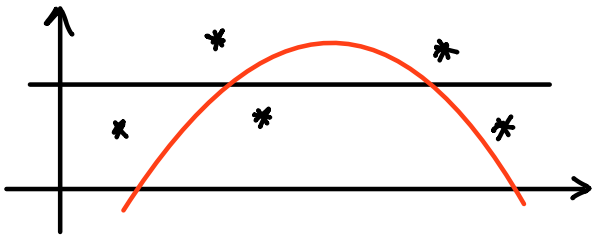
Aproximare în sensul celor mai mici pătrate

$y = f(x)$

y_i ↑ x_i Interpolare

$P_u(x_i) = y_i$

Aproximare (x_i, y_i)



Spațiu prehilbertian (F, u) $\quad u : F \times F \to \mathbb{R}$

F = mulțime funcții

u = produs scalar

Proprietăți produs scalar

a) $\langle f_1 + f_2, f \rangle = \langle f_1, f \rangle + \langle f_2 + f \rangle$

$\langle \alpha f_1, f \rangle = \alpha \langle f_1, f \rangle$

b) $\langle f_1, f_2 \rangle = \langle f_2, f_1 \rangle$

c) $\langle f, f \rangle \geq 0$

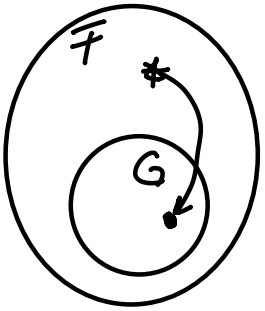
$\langle f, f \rangle = 0 \iff f = 0$

ex $F = \mathbb{R}^3 \qquad \langle x, y \rangle = \sum_{i=1}^{3} x_i y_i$

ex $F = C([a,b]) \qquad \langle f, g \rangle = \int_a^b f(x) \cdot g(x) w(x) dx$

$$\langle f,g\rangle = \sum_{i=1}^{n} f(x_i)g(x_i)w(x_i)$$

$$\|f\| = \sqrt{\langle f,f\rangle}$$



$g^* \in G$ este cel mai bun aproximant în sensul celor mai mici pătrate pt. $f \in F$

$$\|f-g^*\| = \min_{g\in G} \|f-g\| \quad g\in G$$

Th 1

Cond. necesară și suficientă ca $g^* \in G \subset F$ să fie cel mai bun aproximant în sensul celor mai mici pătrate pt. un $f \in F$ este $\langle f-g^*, g\rangle = 0$; $g \in G$

Th 2

g^* este unic

Eroarea

$$E = E_2(a_0,a_1) = \sum_{i=1}^{m} [y_i - (a_1x_i + a_0)]^2$$

Dem → <u>cond. necesară</u> $\langle f-g^*, g\rangle = k \neq 0$

alegem $g_2 = g^* + \frac{k}{\|g_1\|^2} \cdot g_2$ $\quad g_1, g_2, g^* \in G$

$$\|f-g_2\|^2 = \langle f-g_2, f-g_2\rangle$$

$$= \left\langle f-g^* - \frac{k}{\|g_1\|^2}\cdot g_1,\ f-g^* - \frac{k}{\|g_1\|^2}\cdot g_1\right\rangle$$

$$= \langle f-g^*, f-g^*\rangle - 2\left\langle f-g^*, \frac{k}{\|g_1\|^2}\cdot g_1\right\rangle + \left\langle \frac{k}{\|g_1\|^2}\cdot g_1, \frac{k}{\|g_1\|^2}\cdot g_1\right\rangle$$

$$= \|f-g^*\|^2 - 2\cdot\frac{k}{\|g_1\|^2}\underbrace{\langle f-g^*, g_1\rangle}_{k} + \frac{k^2}{\|g_1\|^4}\underbrace{\langle g_1, g_1\rangle}_{\|g_1\|^2}$$

$$\|f-g_2\|^2 = \|f-g^*\|^2 - 2\cdot\frac{k^2}{\|g_1\|^2} + \frac{k^2}{\|g_1\|^2} = \|f-g^*\|^2 - \frac{k^2}{\|g_1\|^2}$$

$$\Rightarrow \|f-g_2\|^2 < \|f-g^*\|^2 \Rightarrow g_2 = g^* \Rightarrow k=0 \quad \text{↯}$$

Deci → cond. suficientă

$$\langle f-g_1, g\rangle = 0 \qquad g_1 \neq g^*$$

$$\|f-g\|^2 = \langle f-g_1, f-g\rangle$$

$$= \langle f-g_1-(g-g_1),\ f-g_1-(g-g_1)\rangle$$

$$= \langle f-g_1, f-g_1\rangle - 2\langle f-g_1, g-g_1\rangle + \langle g-g_1, g-g_1\rangle$$

$$= \|f-g_1\|^2 + \|g-g_1\|^2 - 2\cdot\underbrace{\langle f-g_1, g-g_1\rangle}_{=0}$$

$\|f-g\|^2 > \|f-g_1\|^2 \quad \forall g \in G \Rightarrow g_1 = g^*$

G = spațiu

fie $u_1, u_2 \dots u_n$ o bază pentru G

$$g = \sum_{i=1}^{n} a_i u_i \qquad g^* = \sum_{i=1}^{n} c_i u_i$$

$$\langle f - g^*, g \rangle = 0$$

$$\langle f - g^*, \sum_{j=1}^{n} a_j u_j \rangle = 0 \Rightarrow \sum_{j=1}^{n} a_j \langle f - g^*, u_j \rangle = 0$$

$$\Rightarrow \langle f - g^*, u_j \rangle = 0 \quad j = 1:n$$

$$\langle f - \sum_{i=1}^{n} c_i u_i, u_j \rangle = 0 \Rightarrow \langle f, u_j \rangle - \sum_{i=1}^{n} c_i \langle u_i, u_j \rangle = 0$$

$$\Rightarrow \sum_{i=1}^{n} c_i \langle u_i, u_j \rangle = \langle f, u_j \rangle \quad j = 1:n$$

$\{u_i\}$ - bază ortogonală $\langle u_i, u_j \rangle = \begin{cases} 0 & i \neq j \\ \neq 0 & i = j \end{cases}$

$$c_i = \frac{\langle f, u_i \rangle}{\langle u_i, u_i \rangle} \quad i = 1:n$$

$\{u_i\}$ - ortonormată $\langle u_i, u_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$

$c_i^* = \langle f, u_i \rangle \quad i = 1:n$ coeficienți Fourier

Problemă examen

Determinați cel mai bun aproximat de gr. 2 pt. fct. $f:[-1;1]\to\mathbb{R}$ $f(x)=|x|$; $g_2^*=?$

$$\langle f,g\rangle=\int_{-1}^{1} f(x)g(x)\,dx$$

$$g_2^*(x)=c_0+c_1x+c_2x^2$$

$$\begin{cases} c_0\langle 1,1\rangle+c_1\langle x,1\rangle+c_2\langle x^2,1\rangle=\langle |x|,1\rangle \\ c_0\langle 1,x\rangle+c_1\langle x,x\rangle+c_2\langle x^2,x\rangle=\langle |x|,x\rangle \\ c_0\langle 1,x^2\rangle+c_1\langle x,x^2\rangle+c_2\langle x^2,x^2\rangle=\langle |x|,x^2\rangle \end{cases}$$

$$\langle 1,1\rangle=\int_{-1}^{1}1\,dx=2$$

$$\langle 1,x\rangle=\int_{-1}^{1}x\,dx=\frac{x^2}{2}\Big|_{-1}^{1}=0$$

$$\langle 1,x^2\rangle=\langle x,x\rangle=\int_{-1}^{1}x^2dx=\frac{x^3}{3}\Big|_{-1}^{1}=\frac{2}{3}$$

$$\langle x^2,x\rangle=\int_{-1}^{1}x^3dx=0$$

$$\langle x^2,x^2\rangle=\int_{-1}^{1}x^4dx=\frac{x^5}{5}\Big|_{-1}^{1}=\frac{2}{5}$$

$$\langle |x|, 1\rangle = \int_{-1}^{1} |x|\,dx = 2\int_{0}^{1} x\,dx = 2\cdot \frac{x^2}{2}\Big|_0^1 = 1$$

$$\langle |x|, x\rangle = 0 \qquad \langle |x|, x^2\rangle = \frac{1}{2}$$

$$\begin{cases} 2C_0 + \frac{2}{3}C_2 = 1 \\ \frac{2}{3}C_1 = 0 \\ \frac{2}{3}C_0 + \frac{2}{5}C_2 = \frac{1}{2} \end{cases} \Rightarrow \boxed{C_1 = 0} \Rightarrow \begin{cases} 9C_0 + 3C_2 = \frac{9}{2} \\ 5C_0 + 3C_2 = \frac{15}{4} \end{cases}$$

$$\Rightarrow 4C_0 = \frac{3}{4} \Rightarrow \boxed{C_0 = \frac{3}{16}}$$

$$C_2 = \frac{3}{2} - \frac{9}{16} = \frac{15}{16}$$

$$g_2^*(x) = \frac{15}{16}x^2 + \frac{3}{16}$$

Problemă examen

x	-1	$-1/2$	0	$1/2$	1
f	1	$1/2$	0	$1/2$	1

$$g_2^*(x) = C_0 + C_1 x + C_2 x^2$$

$$C_0\langle 1,1\rangle + C_1\langle x,1\rangle + C_2\langle x^2,1\rangle = \langle f,1\rangle = 3$$

$$C_0\langle 1,x\rangle + C_1\langle x,x\rangle + C_2\langle x^2,x\rangle = \langle f,x\rangle = 0$$

$$C_0\langle 1,x^2\rangle + C_1\langle x,x^2\rangle + C_2\langle x^2,x^2\rangle = \langle f,x^2\rangle = \frac{9}{4}$$

$$\langle 1,1\rangle = \sum_{i=1}^{5} 1 = 5$$

$$\langle 1,x\rangle = \sum_{i=1}^{5} x_i = 0 = \langle x, x^2\rangle = \langle x^2, x\rangle$$

$$\langle 1,x^2\rangle = 2 + 2\cdot\frac{1}{4} = 2 + \frac{1}{2} = \frac{5}{2}$$

$$\langle x^2, x^2 \rangle = \frac{17}{8}$$

$$\begin{cases} 5C_0 + \frac{5}{2}C_2 = 3 \\ \frac{5}{2}C_1 = 0 \longrightarrow \boxed{C_1 = 0} \\ \frac{5}{2}C_0 + \frac{17}{8}C_2 = \frac{9}{4} \end{cases}$$

$$\begin{cases} 2C_0 + C_2 = \frac{6}{5} \\ 20C_0 + 17C_2 = 18 \end{cases}$$

$$\begin{cases} 20C_0 + 10C_2 = 12 \\ 20C_0 + 17C_2 = 18 \end{cases} \Rightarrow 7C_2 = 6 \Rightarrow \boxed{C_2 = \frac{6}{7}}$$

$$C_0 = \frac{1}{2}\left(\frac{6}{5} - \frac{6}{7}\right) = \frac{6}{35} \qquad g_2^*(x) = \frac{6}{7}x^2 + \frac{6}{35}$$

Polinoame Cebîşov

$$T_n(x): [-1,1] \longrightarrow [-1,1] \; ; \; T_n(x) = \cos(n \arccos(x))$$

$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_2(x) = 2x^2 - 1$$

$$T_3(x) = 4x^3 - 3x$$

$$\vdots$$

$$T_{n+1}(x) - 2x\,T_n(x) + T_{n-1}(x) = 0$$

$$\int_{-1}^{1} \frac{T_p(x)\,T_q(x)}{\sqrt{1-x^2}}\,dx = \begin{cases} \pi & p = q = 0 \\ \pi/2 & p = q \neq 0 \\ 0 & p \neq q \end{cases}$$

rădăcini $T_n(x_k)=0 \Rightarrow n\arccos(x_k)=\frac{\pi}{2}+k\pi \quad k=0:n-1$

$$x_k=\cos\left(\frac{2k+1}{2n}\pi\right)$$

extreme $T_n(x_p)=\pm 1 \Rightarrow n\arccos(x_p)=p\pi$

$$\Rightarrow x_p=\cos\left(\frac{p\pi}{n}\right) \qquad p=0:n-1$$

Tipuri de aproximarea în sensul celor mai mici pătrate

(1) continuă

(a) trigonometrică

baza $\left\{\frac{1}{\sqrt{2}}\ \sin x\ \cos x\ \sin 2x\ \cos 2x \dots \sin nx\ \cos nx\right\}$

$$\frac{1}{\pi}\int_0^{2\pi} u_p(x)\,u_q(x)\,dx=\delta_{pq}=\begin{cases}0 & ; p\neq q\\ 1 & ; p=q\end{cases}$$

$$g_n^*(x)=\frac{a_0}{\sqrt{2}}+\sum_{k=1}^{n}\left(a_k\sin kx+b_k\sin kx\right)$$

$$a_0=\left\langle f,\frac{1}{\sqrt{2}}\right\rangle=\frac{1}{\pi\sqrt{2}}\int_0^{2\pi} f(x)\,dx$$

$$a_k = \langle f_1, \sin kx \rangle = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin kx \, dx$$

$$b_k = \langle f_1, \cos kx \rangle = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos kx \, dx$$

Deme

$$f(x) = \sum_{k=1}^{n} (a_k \sin kx + b_k \sin kx)$$

$$f(x) = \frac{1}{n} \sum_{k=0}^{m-1} c_k e^{ikx} \qquad T = \frac{2\pi}{k}$$

c_k = comp. pe frecventa $\frac{k}{2\pi}$ a fct. f_1

$$x_j = -\pi + \frac{j}{m} \pi$$

$$c_k = \sum_{0}^{2m-1} y_j e^{ik\pi j/m}$$

Fast Fourier

$$c_k + c_{k+m} = \sum_{j=0}^{2m-1} y_j e^{ik\pi j/m} (1 + e^{\pi ij})$$

$$= 2 \sum_{j=0}^{m-1} y_{2j} e^{i(2k)\pi j/m} = E_{m-1}$$

$$c_k - c_{k+m} = 2 e^{ik\pi/m} \left(\sum_{j=0}^{m-1} y_{2j+1} e^{i(2j+1)\pi k/m} \right) = O_{m-1}$$

(b) Polinoame Cebîșev

baza $\left\{\frac{T_0(x)}{\sqrt{2}} \quad T_1(x) \quad T_2(x) \quad \dots \quad T_n(x)\right\}$

$$\int_{-1}^{1} \frac{T_p(x)T_q(x)}{\sqrt{1-x^2}}\,dx = \begin{cases} \pi & p=q=0 \\ \pi/2 & p=q\neq 0 \\ 0 & p\neq q \end{cases}$$

$$g_n^*(x) = \frac{a_0 T_0(x)}{\sqrt{2}} + \sum_{k=1}^{n} a_k T_k(x)$$

$$a_0 = \langle f, \frac{T_0(x)}{\sqrt{2}} \rangle = \frac{1}{\pi\sqrt{2}} \int_{-1}^{1} \frac{f(x)}{\sqrt{1-x^2}}\,dx$$

$$a_k = \langle f, T_k(x) \rangle = \frac{2}{\pi} \int_{-1}^{1} \frac{f(x)\cdot T_k(x)}{\sqrt{1-x^2}}\,dx$$

(2) discretă

(a) trigonometrică

baza $\left\{\frac{1}{\sqrt{2}} \sin x \cos x \sin 2x \cos 2x \dots \sin nx \cos nx\right\}$

$$\sum_{k=1}^{2n+2} u_p(x_i)u_q(x_i) = \begin{cases} 0 \,; & p\neq q \\ n+1 \,; & p=q \end{cases} \qquad x_i = \frac{2k\pi}{2n+2}$$

$$g_n^*(x) = \frac{a_0}{\sqrt{2}} + \sum_{k=1}^{n} (a_k \sin kx + b_k \sin kx)$$

$$a_0 = \frac{1}{\sqrt{2}} \sum_{i=0}^{2u+2} f(x_i)$$

$$a_k = \sum_{i=0}^{2u+2} f(x_i) \sin kx_i$$

$$b_k = \sum_{i=0}^{2u+2} f(x_i) \cos kx_i$$

$f(x_i) = f_i$
valoare măsurată

(b) Polinoame Cebîșev

baza $\left\{ \frac{T_0(x)}{\sqrt{2}} \quad T_1(x) \quad T_2(x) \quad \dots \quad T_u(x) \right\}$

$$\sum_{k=0}^{u} T_p(x_i) T_q(x_i) = \begin{cases} 0 & p \neq q \\ \frac{u+1}{2} & p = q \end{cases}$$

$x_i = \cos\left(\frac{2k+1}{2u+2}\pi\right)$
x_i rad $T_{k+1}(x)$

$$g_u^*(x) = \frac{a_0 T_0(x)}{\sqrt{2}} + \sum_{k=1}^{u} a_k T_k(x)$$

$$a_0 = \frac{\sqrt{2}}{u+1} \sum_{i=0}^{u} f(x_i)$$

$$a_k = \frac{2}{u+1} \sum_{i=0}^{u} f(x_i) T_k(x_i)$$